

Statistics (1)

Syllabus

Yu-You Liou

Fall 2026

Lecturer: Yu-You Liou

Course No.: EIB-21E-01-A1

Meeting Time: Wednesday, Periods 6–7 (13:10–15:00)

Contact: d10627008@ntu.edu.tw

Website: <https://yyliou.github.io/stat>

Location: A207

Course Objectives

1. To train students in basic statistical knowledge and econometric concepts.
2. To enhance students' familiarity with data analysis and provide hands-on experience in statistical analysis.

Textbooks

1. Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **In-Class Exercises** (36%): 12 exercises $\times$ 3% each, submitted at the end of each session.
2. **Mid-Term Exam** (32%) and **Final Exam** (32%).

Regulations

1. Attendance

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. In-class exercise deadline

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

3. Academic Integrity

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

4. AI Policy

Students may use AI tools (e.g., ChatGPT or Gemini) as learning aids for in-class exercises, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty. AI tools are not permitted during examinations.

Exam Regulations

The examination is conducted online under the following regulations.

1. Mandatory Attendance

Students must take the exam in person. Roll call will be taken five minutes before the exam begins. Late arrival or absence will result in a score of zero.

2. Network Restriction

The exam system tracks IP addresses. You must use the campus Wi-Fi. Access from a non-campus IP address will result in a score of zero.

3. One Device Only

The exam must be taken on a single device. Keeping any additional device, such as a phone or a tablet, available during the exam will result in a score of zero.

4. Authorized Resources

This is an open-book exam. Printed reference materials are permitted without limit. Electronic reference materials and AI tools, such as ChatGPT or Gemini, are prohibited. Any use will result in a score of zero.

5. Media Prohibition

Photography, video or audio recording, and screenshotting are strictly prohibited. Violators will receive a score of zero.

6. No Communication or Disturbance

Interpersonal communication, causing noise or disruptive movements are strictly prohibited. Violators will receive a score of zero.

7. Copy-Paste Disabled

The exam system prohibits copying and pasting. Do not attempt to bypass this feature.

Required Software

All software is free and open-source: **R** at <https://cran.r-project.org>, **RStudio Desktop** at <https://posit.co/download/rstudio-desktop>, and R packages **tidyverse**, **ggplot2**, **patchwork** (installation instructions on the course R page).

Course Schedule

Week	Topic
1	Explain the syllabus and class regulations
2	Chapter 1: The nature of probability and statistics (ex1)
3	Chapter 2: Frequency distributions and graphs (ex2)
4	Chapter 3: Data description (ex3)
5	Chapter 3: Data description (ex4)
6	Chapter 4: Probability and counting rules (ex5)
7	Chapter 4: Probability and counting rules (ex6)
8	Midterm review
9	Midterm exam
10	Chapter 5: Discrete probability distributions (ex7)
11	Chapter 5: Discrete probability distributions (ex8)
12	Chapter 6: The normal distribution (ex9)

Week	Topic
13	Chapter 6: The normal distribution (ex10)
14	Chapter 7: Confidence intervals and sample size (ex11)
15	Chapter 7: Confidence intervals and sample size (ex12)
16	Final exam
17	Flexible learning week
18	Flexible learning week

Note: Syllabus is subject to change. Any updates will be announced on the course website.